

Module 2: Data Preprocessing

Types of Data, Sampling Theory, Sampling Techniques, Correlation, Feature Selection. Dimensionality Reduction Techniques: Projections, Eigen Value Decomposition, Principal Component Analysis (PCA).

2.1 Types of Data

Data is the collection of raw facts, measurements, or observations used for analysis, decision-making, and machine learning model building.

2.1.1 Based on Nature of Values (Statistical Measurement Scales)

2.1.1.1 Qualitative (Categorical) Data:

- Represents **categories, labels, or qualities** instead of numbers.
- Cannot be directly measured or subjected to arithmetic operations.

Types:

Nominal Data

- Purely categorical, no order or ranking.
- Numbers may be used as labels, but they don't imply magnitude.

Examples:

Gender → Male, Female, Other

Blood Group → A, B, AB, O

Colors → Red, Blue, Green

Ordinal Data

- Categorical data with a **natural order or ranking**.
- The **intervals between ranks are not equal or meaningful**.

Examples:

- ✓ Education → High School < Graduate < Postgraduate
- ✓ Customer Satisfaction → Poor < Average < Good < Excellent
- ✓ Movie Ratings → ★, ★★, ★★★, ★★★★, ★★★★★

Quantitative (Numerical) Data

Represents **measurable quantities**.

Arithmetic operations (mean, variance, etc.) are valid.

Types

Discrete Data

- ✓ Countable values, integers only.
- ✓ No intermediate values between two points.

Examples:

- ✓ Number of students in a class
- ✓ Number of cars in a parking lot
- ✓ Goals scored in a football match

Continuous Data

- ✓ Can take **any value within a range**, including decimals/fractions.

- ✓ Usually comes from measurement.

Examples:

- ✓ Height (175.4 cm)
- ✓ Weight (65.8 kg)
- ✓ Temperature (36.7 °C)
- ✓ Time (12.58 seconds)

Based on Structure **Structured Data**

- ✓ Well-organized, stored in tables (rows & columns).
- ✓ Easy to search and analyze.

Example: Relational databases, Excel spreadsheets.

Unstructured Data

- ✓ No predefined organization.
- ✓ Difficult to process with traditional methods; requires NLP, CV, etc.

Example: Emails, Images, Videos, Social media posts.

Semi-Structured Data

- ✓ Partially structured, but not in strict table form.
- ✓ Has tags or markers to separate data elements.

Example: XML files, JSON data, NoSQL databases.

Based on Time of Collection

Cross-Sectional Data

- ✓ Collected at **a single point in time**.
- ✓ Provides a snapshot of variables.

Example: Census 2021 population data, income survey at one time.

Time-Series Data

- ✓ Collected **sequentially over time**.
- ✓ Used for trend analysis, forecasting.

Example: Daily stock prices, Monthly rainfall, Yearly GDP growth.

2.1.2 Other Classifications in ML

Context Labeled Data:

Input data with target/output labels (used in supervised learning).

Example: Images with “cat” or “dog” tags.

Unlabeled Data:

No output labels (used in unsupervised learning).

Example: Raw customer transaction logs.

Balanced Data:

Classes are evenly distributed.

Imbalanced Data:

One class heavily outnumbered others (e.g., fraud detection).

2.2 Sampling Theory

Sampling is a method that allows us to get information about the population based on the statistics from a subset of the population (sample), without having to investigate every individual.

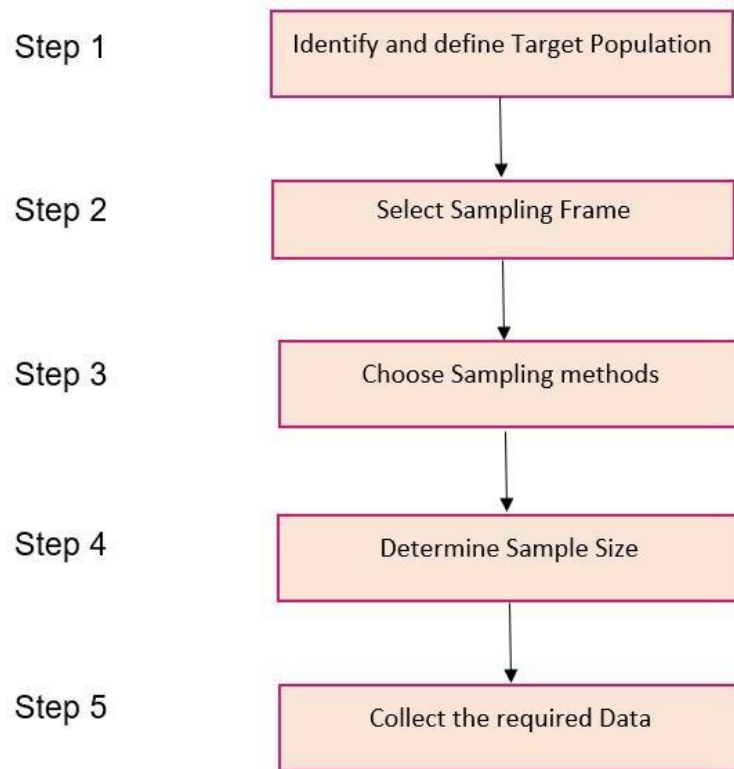


2.2.1 Need of Sampling:

Sampling is done to draw conclusions about populations from samples, and it enables us to determine a population's characteristics by directly observing only a portion (or sample) of the population.

- Selecting a sample requires less time than selecting every item in a population
- Sample selection is a cost-efficient method
- Analysis of the sample is less cumbersome and more practical than an analysis of the entire population.

Steps involved in Sampling:



Let's take an interesting case study and apply these steps to perform sampling. We recently conducted General Elections in India a few months back. You must have seen the public opinion polls every news channel was running at the time:

Step 1:

The first stage in the sampling process is to clearly define the target population.

Step 2:

Sampling Frame : It is a list of items or people forming a population from which the sample is taken. So, the sampling frame would be the list of all the people whose names appear on the voter list of a constituency.

Step 3:

Generally, probability sampling methods are used because every vote has equal value and any person can be included in the sample irrespective of his caste, community, or religion. Different samples are taken from different regions all over the country.

Step 4:

Sample Size : It is the number of individuals or items to be taken in a sample that would be enough to make inferences about the population with the desired level of accuracy and precision. Larger the sample size, more accurate our inference about the population would be. For the polls, agencies try to get as many people as possible of diverse backgrounds to be included in the sample as it would help in predicting the number of seats a political party can win.

Step 5:

Once the target population, sampling frame, sampling technique, and sample size have been established, the next step is to collect data from the sample. In opinion polls, agencies generally put questions to the people, like which political party are they going to vote for or has the previous party done any work, etc. Based on the answers, agencies try to interpret who the people of a constituency are going to vote for and approximately how many seats is a political party going to win.

2.3 Sampling Techniques

Sampling techniques are methods used to select a portion of a population to study, in order to make inferences about the entire population. Broadly, they are divided into two types: probability sampling and non-probability sampling. In probability sampling, every member of the population has a known and non-zero chance of being selected. Common methods include simple random sampling, where every member has an equal chance of selection; systematic sampling, where every k-th member is chosen after a random start; stratified sampling, where the population is divided into homogeneous groups (strata) and samples are taken from each;

cluster sampling, where entire clusters are randomly selected; and multi-stage sampling, which combines several sampling stages.

In non-probability sampling, not every individual has a chance of being selected, often making it less representative but more practical in certain situations. Examples include convenience sampling, where easily available subjects are selected; judgmental or purposive sampling, based on the researcher's judgment; quota sampling, ensuring certain groups are represented in fixed proportions; and snowball sampling, where existing participants recruit others, often used for hard-to-reach populations. Probability sampling is preferred for generalizable, unbiased results, while non-probability sampling is often used in exploratory or qualitative research where probability-based methods are not feasible.

2.3.1 Different Types of Sampling:

The different types of sampling techniques are,

Probability Sampling: In probability sampling, every element of the population has an equal chance of being selected. Probability sampling gives us the best chance to create a sample that is truly representative of the population.

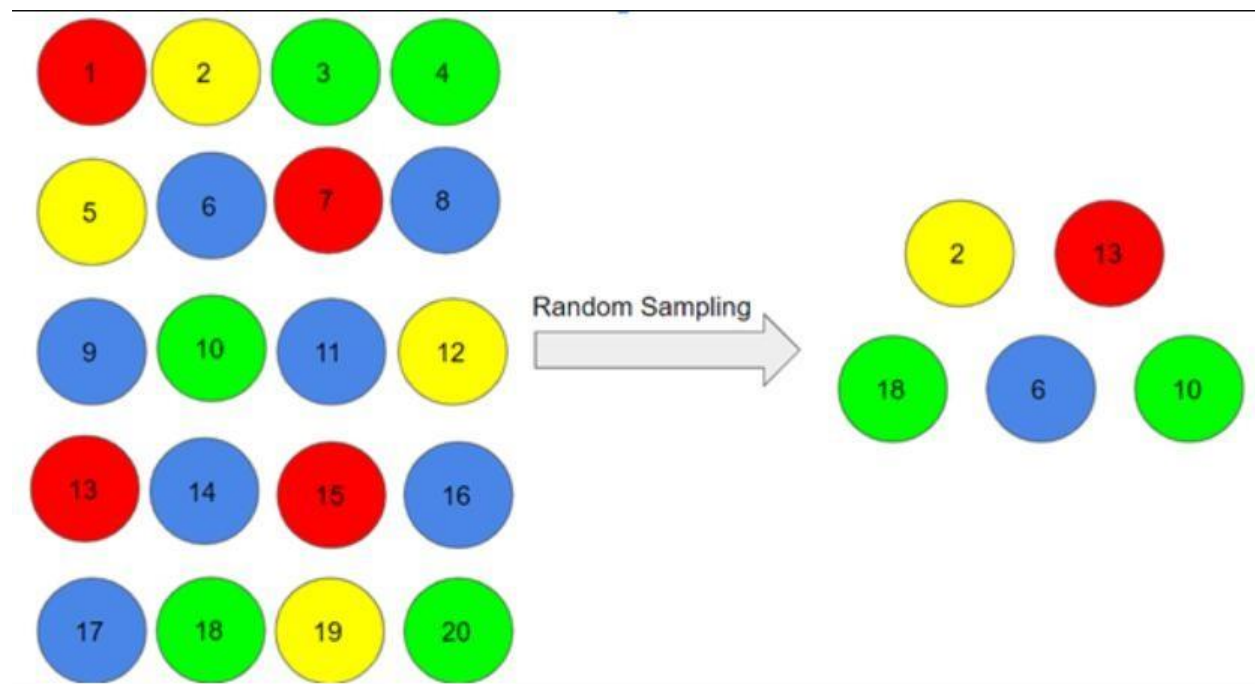
Non-Probability Sampling: In non-probability sampling, all elements do not have an equal chance of being selected. Consequently, there is a significant risk of ending up with a non-representative sample which does not produce generalizable results. For example, let's say our population consists of 20 individuals. Each individual is numbered from 1 to 20 and is represented by a specific color (red, blue, green, or yellow). Each person would have odds of 1 out of 20 of being chosen in probability sampling.

With non-probability sampling, these odds are not equal. A person might have a better chance of being chosen than others. So now that we have an idea of these two sampling types, let's dive into each and understand the different types of sampling under each section.

2.3.2 Types of Probability Sampling:

2.3.2.1 Simple Random Sampling:

This is a type of sampling technique you must have come across at some point. Here, every individual is chosen entirely by chance and each member of the population has an equal chance of being selected. Simple random sampling reduces selection bias.

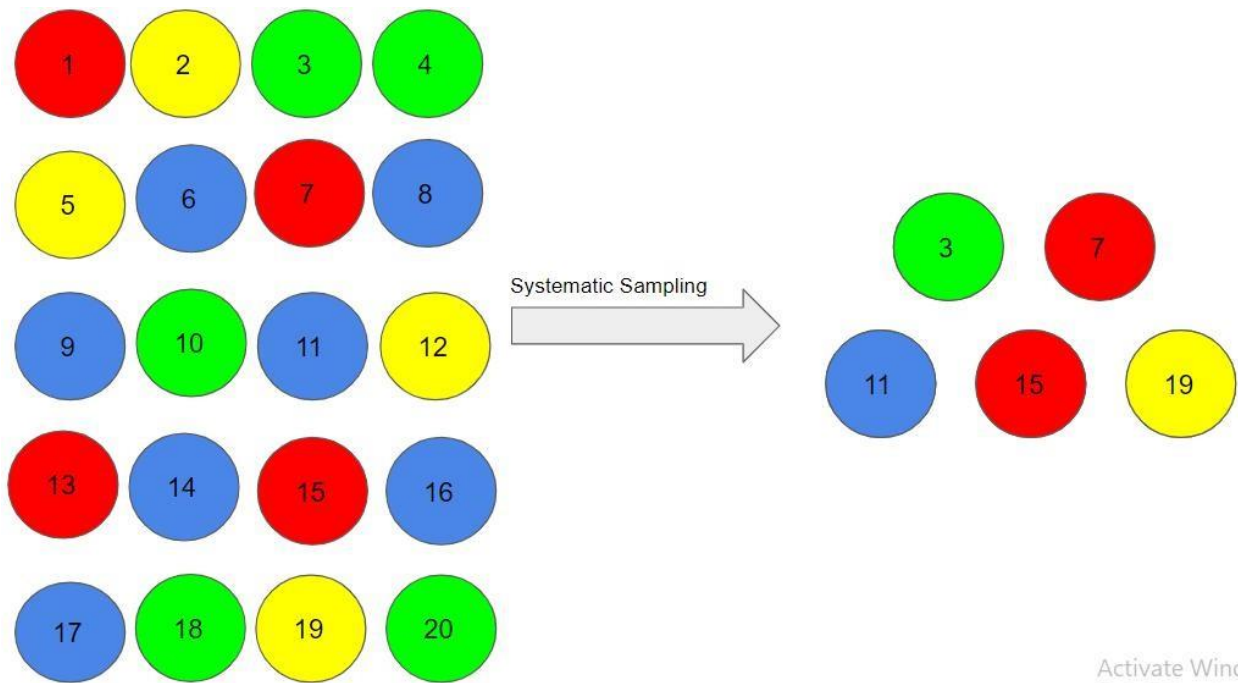


One big advantage of this technique is that it is the most direct method of probability sampling. But it comes with a caveat – it may not select enough individuals with our characteristics of interest. Monte Carlo methods use repeated random sampling for the estimation of unknown parameters.

2.3.2.2 Systematic Sampling:

In this type of sampling, the first individual is selected randomly and others are selected using a fixed ‘sampling interval’. Let’s take a simple example to understand this.

Say our population size is x and we have to select a sample size of n . Then, the next individual that we will select would be x/n th intervals away from the first individual. We can select the rest in the same way.



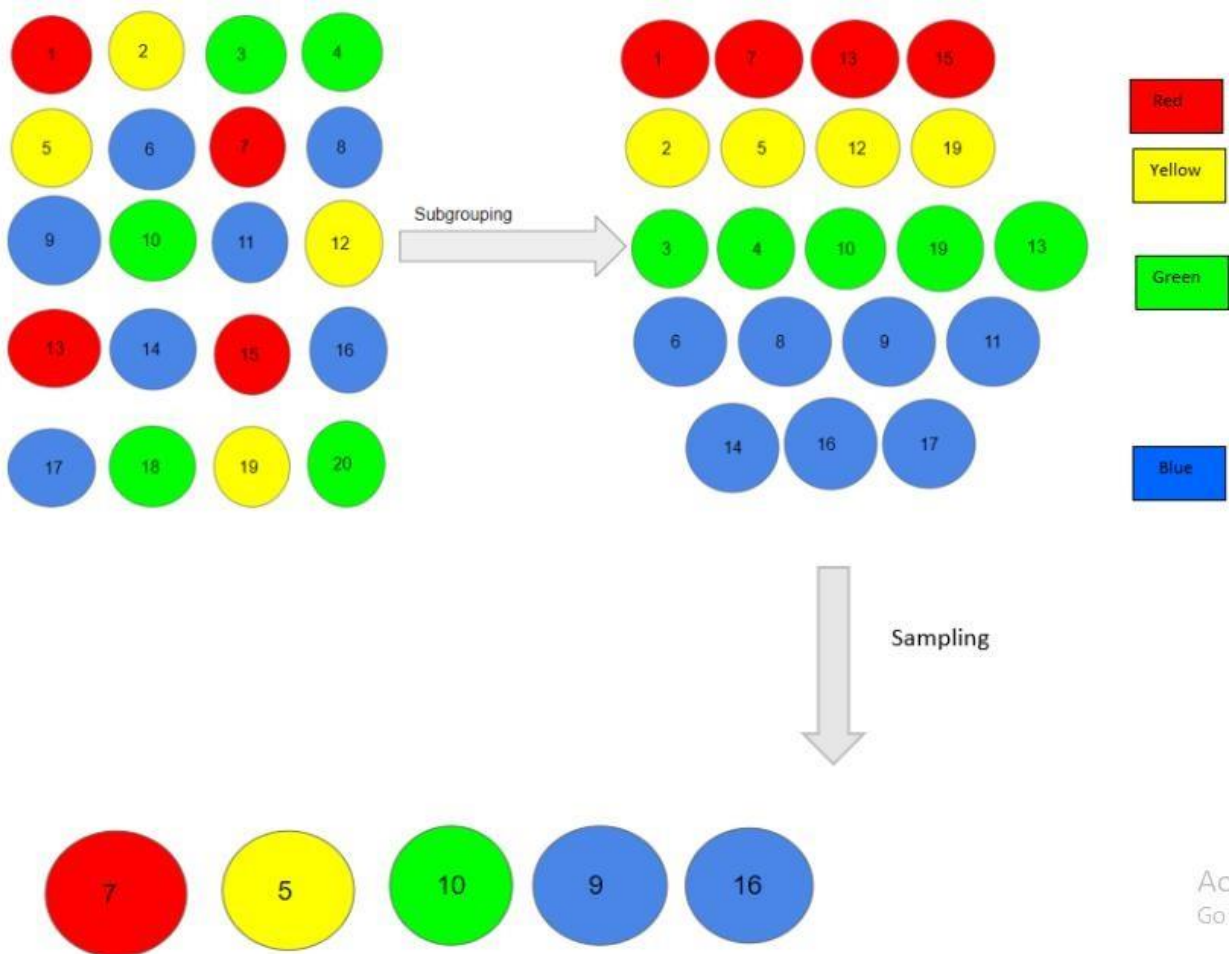
Suppose, we began with person number 3, and we want a sample size of 5. So, the next individual that we will select would be at an interval of $(20/5) = 4$ from the 3rd person, i.e. 7 (3+4), and so on.

3, $3+4=7$, $7+4=11$, $11+4=15$, $15+4=19$ = 3, 7, 11, 15, 19

Systematic sampling is more convenient than simple random sampling. However, it might also lead to bias if there is an underlying pattern in which we are selecting items from the population (though the chances of that happening are quite rare).

2.3.2.3 Stratified Sampling:

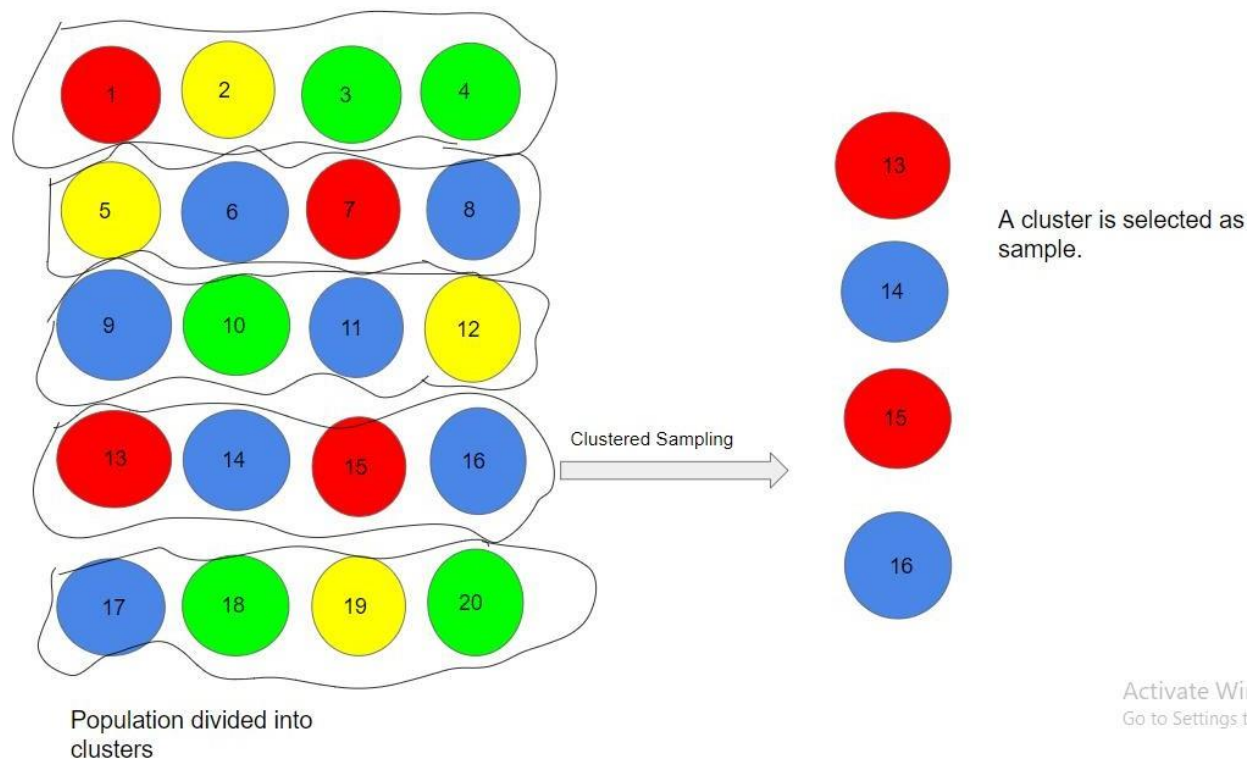
In this type of sampling, we divide the population into subgroups (called strata) based on different traits like gender, category, etc. And then we select the sample(s) from these subgroups:



Here, we first divided our population into subgroups based on different colors of red, yellow, green and blue. Then, from each color, we selected an individual in the proportion of their numbers in the population. We use this type of sampling when we want representation from all the subgroups of the population. However, stratified sampling requires proper knowledge of the characteristics of the population.

2.3.2.4 Cluster Sampling:

In a clustered sample, we use the subgroups of the population as the sampling unit rather than individuals. The population is divided into subgroups, known as clusters, and a whole cluster is randomly selected to be included in the study:



In the above example, we have divided our population into 5 clusters. Each cluster consists of 4 individuals and we have taken the 4th cluster in our sample. We can include more clusters as per our sample size.

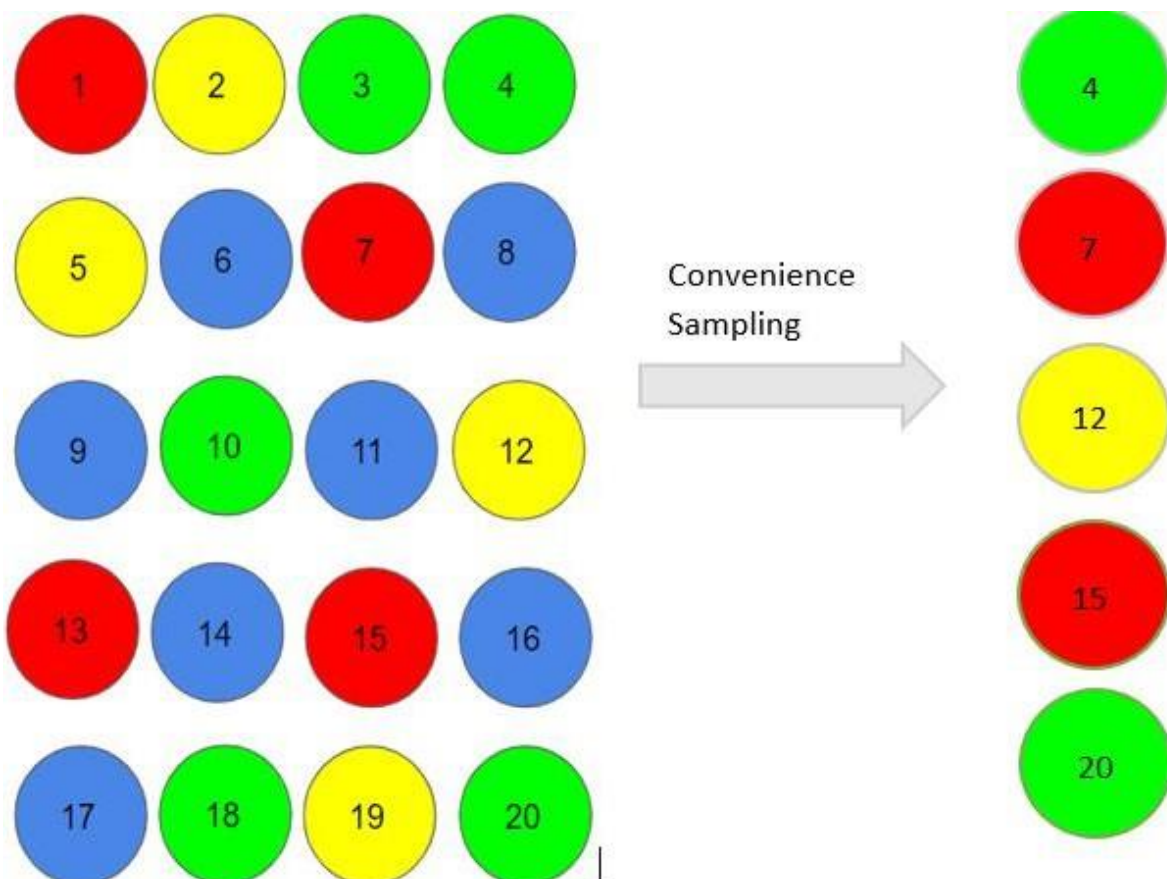
This type of sampling is used when we focus on a specific region or area.

2.3.3 Types of Non-Probability Sampling:

2.3.3.1 Convenience Sampling:

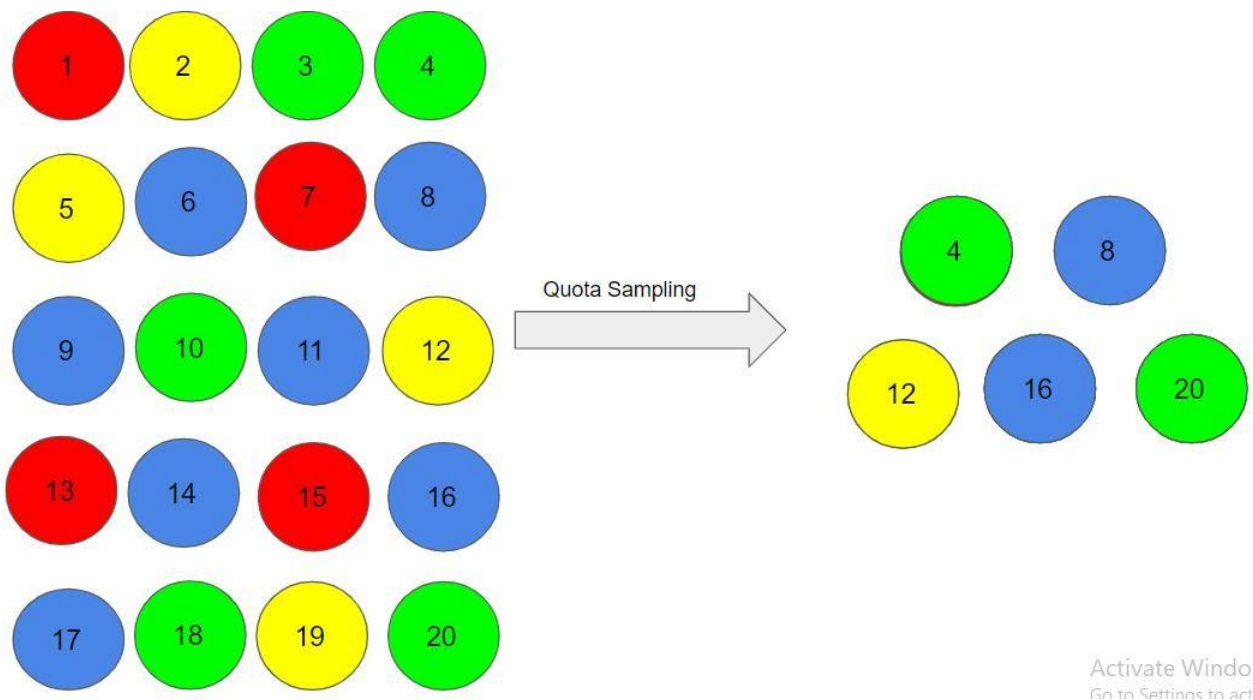
This is perhaps the easiest method of sampling because individuals are selected based on their availability and willingness to take part.

Here, let's say individuals numbered 4, 7, 12, 15 and 20 want to be part of our sample, and hence, we will include them in the sample.



Convenience sampling is prone to significant bias, because the sample may not be the representation of the specific characteristics such as religion or, say the gender, of the population.

2.3.3.2 Quota Sampling:



In this type of sampling, we choose items based on predetermined characteristics of the population.

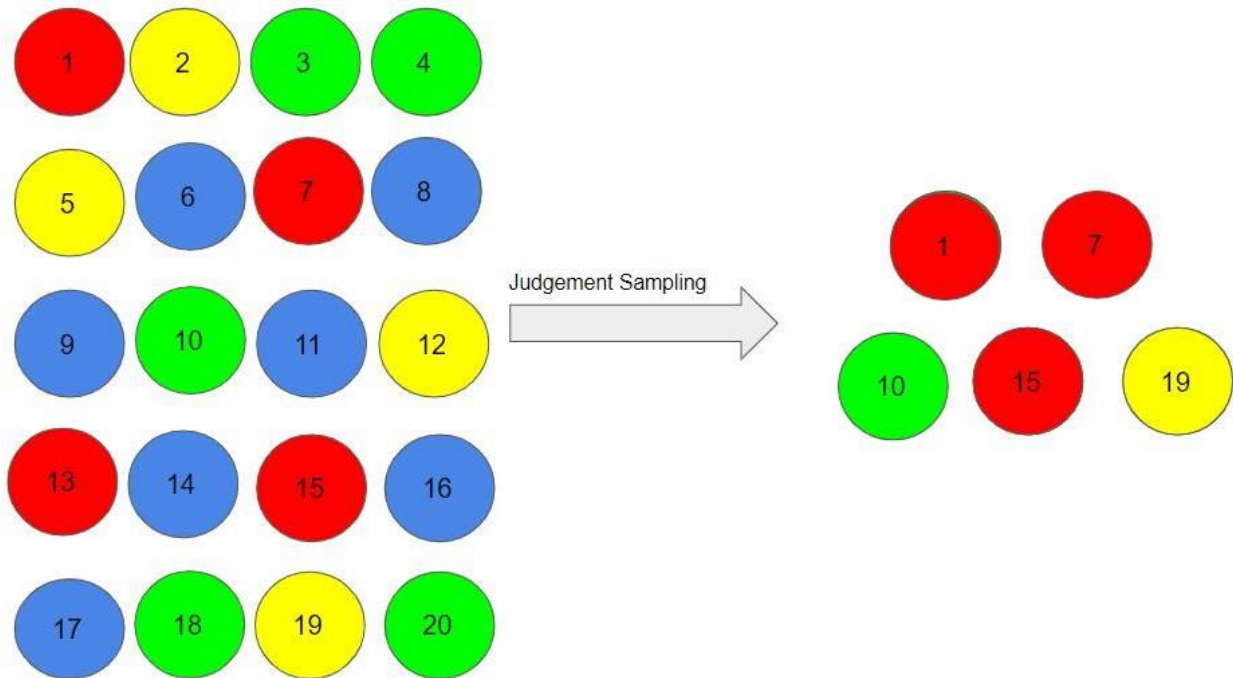
Consider that we have to select individuals having a number in multiples of four for our sample:

Therefore, the individuals numbered 4, 8, 12, 16, and 20 are already reserved for our sample.

In quota sampling, the chosen sample might not be the best representation of the characteristics of the population that weren't considered.

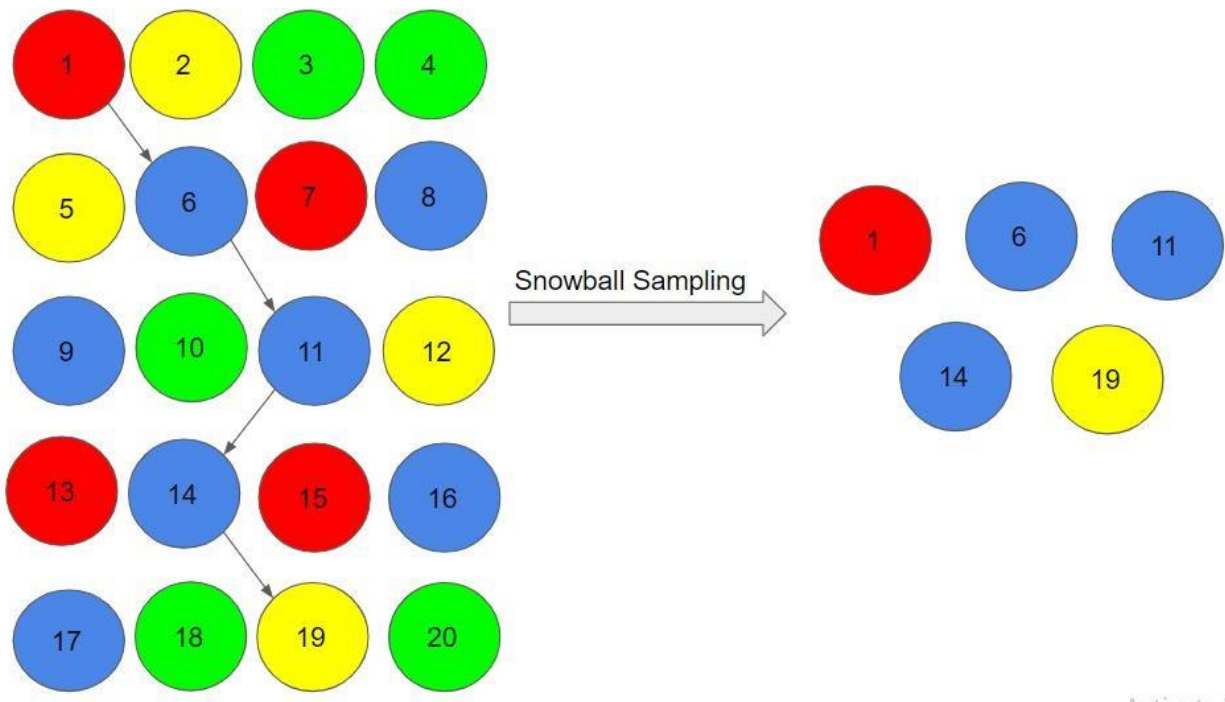
2.3.3.3 Judgment Sampling:

It is also known as selective sampling. It depends on the judgment of the experts when choosing whom to ask to participate.



Suppose, our experts believe that people numbered 1, 7, 10, 15, and 19 should be considered for our sample as they may help us to infer the population in a better way. As you can imagine, quota sampling is also prone to bias by the experts and may not necessarily be representative.

2.3.3.4 Snowball Sampling:



I quite like this sampling technique. Existing people are asked to nominate further people known to them so that the sample increases in size like a rolling snowball. This method of sampling is effective when a sampling frame is difficult to identify.

Here, we had randomly chosen person 1 for our sample, and then he/she recommended person 6, and person 6 recommended person 11, and so on.

1->6->11->14->19

There is a significant risk of selection bias in snowball sampling, as the referenced individuals will share common traits with the person who recommends them.

Example:

A city has **9 lakhs registered voters**.

A political research team wants to conduct a **survey on voter preferences before elections**.

They have limited **budget and time**.

They consider the following options for **sampling**:

(a) Simple Random Sampling – Generate a list of all voters and randomly pick 1000 voters.

(b) Stratified Sampling – Divide voters into strata based on **age groups: 18–30, 31–50, 51+**, and then randomly sample **400, 400, and 200** from each group.

(c) Cluster Sampling – Randomly select 10 wards in the city and survey all voters in those wards.

(d) Convenience (N-P) – Stand near a busy metro station and ask 1000 passers-by about their voting preferences.

(e) Snowball Sampling – Start with 10 known voters and ask each of them to refer more voters until 1000 are surveyed.

Questions

(i) Which method ensures the highest accuracy of representing the entire city?

→ **Stratified or Simple Random Sampling**

(ii) Which is the cheapest and fastest?

→ **Convenience Sampling**

(iii) If the team specifically wants to understand the opinion of young voters (18–30), which method is suitable?

→ **Stratified Sampling**

(iv) Why does cluster sampling give misleading results in this case?

→ Because selected clusters (wards) may not represent the entire city accurately.

(v) If a survey is about a sensitive issue, which method can be more practical?

→ **Snowball Sampling**

2.4 Correlation

Correlation: The degree of relationship between the variables under consideration is measure through the correlation analysis.

- The measure of correlation called the correlation coefficient
- The degree of relationship is expressed by coefficient which range from correlation $(-1 \leq r \leq +1)$
- The direction of change is indicated by a sign.
- The correlation analysis enable us to have an idea about the degree & direction of the relationship between the two variables under study.

Correlation is a statistical tool that helps to measure and analyze the degree of relationship between two variables.

Correlation analysis deals with the association between two or more variables.

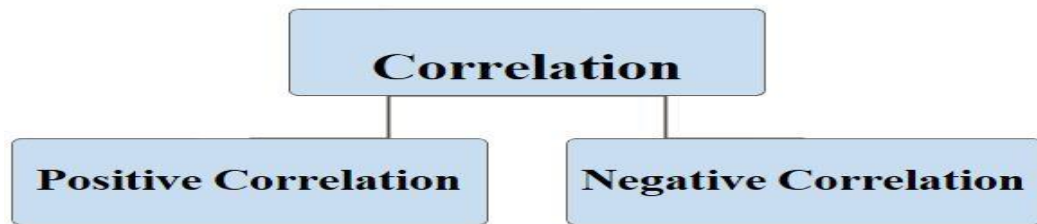
2.4.1 Correlation:

Causation means cause & effect relation. Correlation denotes the interdependency among the variables for correlating two phenomenon, it is essential that the two phenomenon should have cause-effect relationship,& if such relationship does not exist then the two phenomenon can not be correlated.

If two variables vary in such a way that movement in one are accompanied by movement in other, these variables are called cause and effect relationship.

Causation always implies correlation but correlation does not necessarily implies causation.

2.4.2 Types of Correlation Type-I:



Types of Correlation Type-I:

Positive Correlation: The correlation is said to be positive correlation if the values of two variables changing with same direction.

Ex. Pub. Exp. & sales, Height & weight.

Negative Correlation: The correlation is said to be negative correlation when the values of variables change with opposite direction.

Ex. Price & qty. demanded.

Direction of the Correlation:

Positive relationship – Variables change in the same direction.

(Indicated by sign; (+) or (-).)

As X is increasing, Y is increasing

As X is decreasing, Y is

decreasing

E.g., As height increases, so does weight.

Negative relationship – Variables change in opposite directions.

As X is increasing, Y is decreasing

As X is decreasing, Y is increasing

E. g., As TV time increases, grades decrease

More examples:

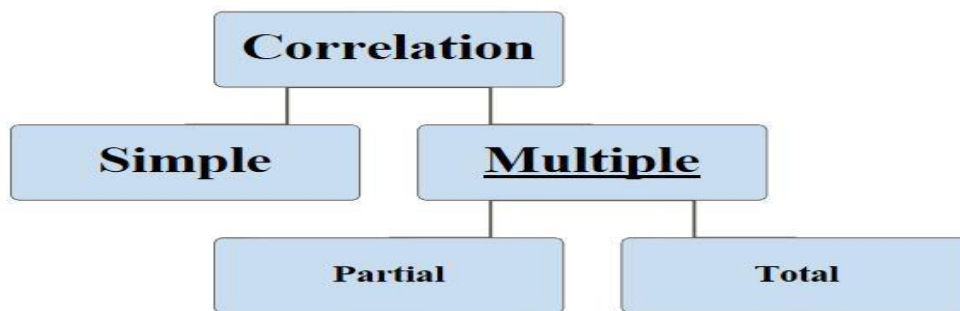
Positive relationships:

- ✓ water consumption and temperature.
- ✓ study time and grades.

Negative relationships:

- ✓ alcohol consumption and driving ability.
- ✓ Price & quantity demanded

2.4.3 Types of Correlation Type-II:



Types of Correlation Type-II

Simple correlation: Under simple correlation problem there are only two variables are studied.

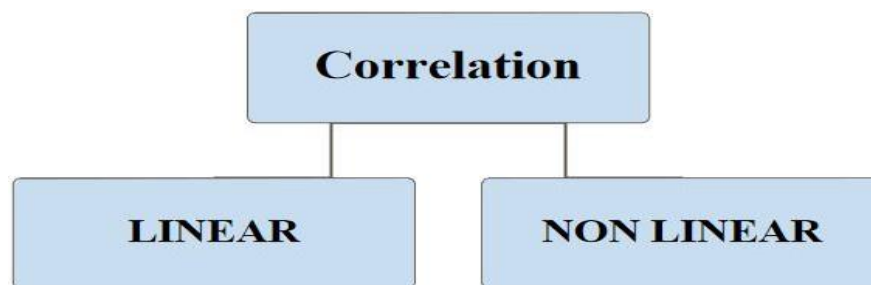
Multiple Correlation: Under Multiple Correlation three or more than three variables are studied.

Ex. $Q_d = f(P, P_C, P_S, t, y)$

Partial correlation: analysis recognizes more than two variables but considers only two variables keeping the other constant.

Total correlation: is based on all the relevant variables, which is normally not feasible.

2.4.4 Types of Correlation Type-III:



Types of Correlation Type-III

Linear correlation: Correlation is said to be linear when the amount of change in one variable tends to bear a constant ratio to the amount of change in the other. The graph of the variables having a linear relationship will form a straight line.

Ex $X = 1, 2, 3, 4, 5, 6, 7, 8,$

$Y = 5, 7, 9, 11, 13, 15, 17, 19,$

$$Y = 3 + 2x$$

Non Linear correlation: The correlation would be non linear if the amount of change in one variable does not bear a constant ratio to the amount of change in the other variable.

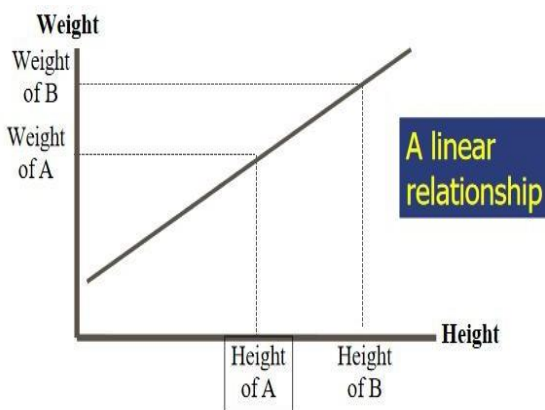
2.4.5 Methods of Studying Correlation:

- ✧ Scatter Diagram Method
- ✧ Karl Pearson's Coefficient of Correlation

Scatter Diagram Method:

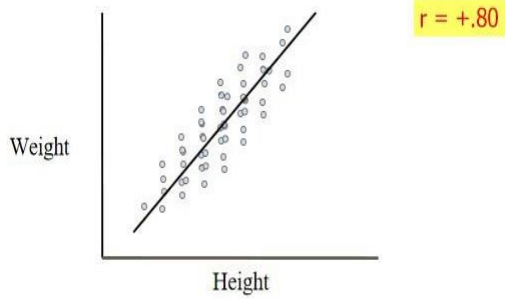
Scatter Diagram is a graph of observed plotted points where each point represents the values of X & Y as a coordinate. It portrays the relationship between these two variables graphically.

A perfect positive correlation

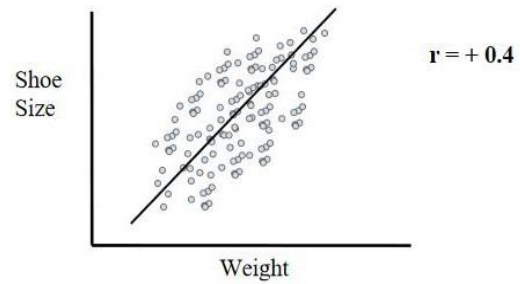


High Degree of positive correlation:

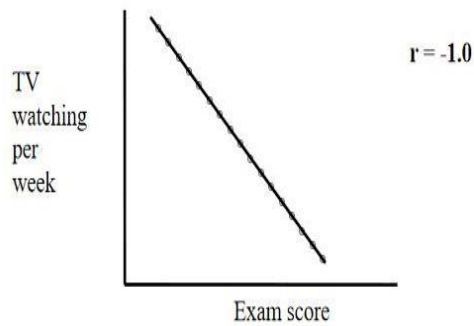
■ Positive relationship



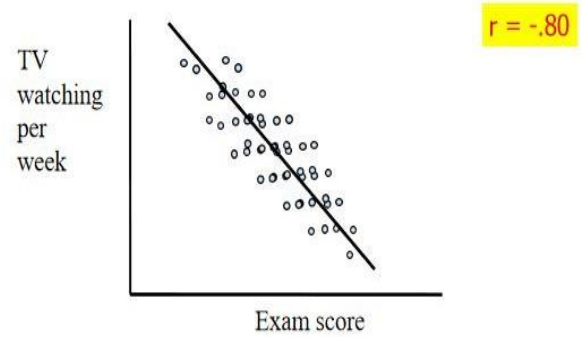
■ Moderate Positive Correlation



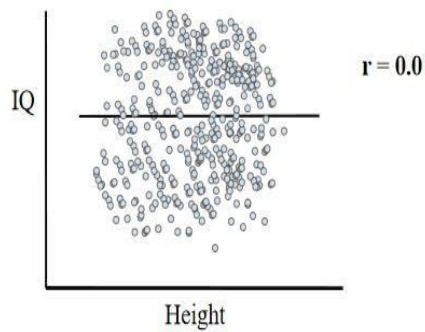
■ Perfect Negative Correlation



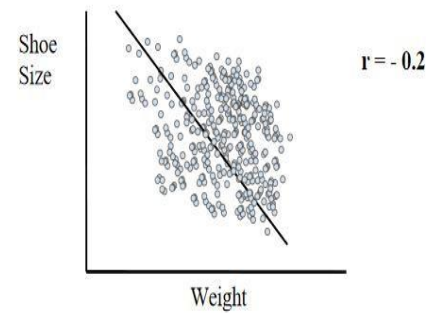
■ Moderate Negative Correlation



■ No Correlation (horizontal line)



■ Weak negative Correlation



Advantages of Scatter Diagram:

- ✧ Simple & Non Mathematical method
- ✧ Not influenced by the size of extreme item
- ✧ First step in investigating the relationship between two variables
- ✧ Disadvantage of scatter diagram:
- ✧ Can not adopt the an exact degree of correlation

Karl Pearson's Coefficient of Correlation:

Method:- 2 Karl Pearson Coefficient

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}$$

This gives the coefficient value which is in the range (-1 to +1) using above formula.

1. Calculate the Karl Pearson Coefficient & interpret the result.

x	Hours Studied	2	4	6	8	10
y	Result	50	60	65	70	80

$\bar{x} = 6$
 $\bar{y} = 65$

$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
$(2-6) = -4 = 16$	$(50-65) = -15 = 225$	60
$(4-6) = -2 = 4$	$(60-65) = -5 = 25$	10
$(6-6) = 0 = 0$	$(65-65) = 0 = 0$	0
$(8-6) = 2 = 4$	$(70-65) = 5 = 25$	10
$(10-6) = 4 = 16$	$(80-65) = 15 = 225$	60
		<u>140</u>

$$r = \frac{140}{40(500)} = \frac{140}{\sqrt{2000}} = \frac{140}{44.721} = 3.13$$

$r = +0.99 \Rightarrow \text{very strong}$

2.	x	1	2	3	4	5
	y	90	85	75	65	60

x	y	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
1	90	-2 = 4	15 = 225 - 30	
2	85	-1 = 1	10 = 100 - 10	
3	75	0 = 0	0	0
4	65	1 = 1	-10 = 100 - 10	
5	60	2 = 4	-15 = 225 - 35	
				-80

$\bar{x} = 3$
 $\bar{y} = 75$

$$r = \frac{-80}{10(650)} = \frac{-80}{\sqrt{6500}} = -0.99$$

Strong negative Correlation.

Advantages of Pearson's Coefficient:

- ✓ It summarizes in one value, the degree of correlation & direction of correlation also.
- ✓ Limitation of Pearson's Coefficient:
- ✓ Always assume linear relationship
- ✓ Interpreting the value of r is difficult.
- ✓ Value of Correlation Coefficient is affected by the extreme values.

2.5 Feature Selection

Feature Selection = the process of selecting the most **relevant features (independent variables)** from your dataset to improve model performance.

- Why needed? → To reduce **overfitting, training time, noise**, and improve **accuracy**.
- Example: If predicting house prices, features like *location, size, number of rooms* matter, but *color of the door* might not.

Supervised vs Unsupervised Feature Selection

1. Supervised Feature Selection

- Uses **target labels (Y)** to guide which features are important.
- Example:
 - Dataset → Student performance with features: *study_hours, attendance, hobbies, age, marks*.
 - Since the label (*marks*) is available, we can test which features strongly correlate with marks (*study_hours, attendance*) → keep those.

2. Unsupervised Feature Selection

- No labels (Y). Focus is only on **relationships among features**.
- Example:
 - Dataset → Customer details without purchase info.
 - If *age* and *income* are highly correlated (redundant), we keep one of them.

Feature Selection Methods

1. Filter Methods

Use statistical measures to rank or select features.

Method	Description
Correlation Coefficient	For numeric features — keep features highly correlated with target
Chi-square Test	For categorical features — measures independence between feature and target
Information Gain	Based on entropy; measures how much information a feature gives about target

2. Wrapper Methods

Use model performance to evaluate subsets of features.

Method	Description
Forward Selection	Start with no features → add one at a time that improves performance

Method	Description
Backward Elimination	Start with all features → remove least significant features step by step
Recursive Feature Elimination (RFE)	Train model → rank features → remove weakest recursively

3. Embedded Methods

Feature selection happens during model training.

Model	How it selects
Lasso Regression	Shrinks less important coefficients to zero
Decision Tree	Uses information gain for splitting, giving feature importance
Random Forest	Gives average feature importance from multiple trees

Filter Method

• Correlation Coefficient

```
import pandas as pd
data = {'Study hrs': [2,4,6,8,10],
        'Attendance': [50,60,70,80,90],
        'Age': [15,15.5,15,15,15],
        'Marks': [60,70,80,90,95]}
```

```
df = pd.DataFrame(data)
print(df.corr()['Marks'])
```

Sample Output:

- Study hours to marks → 0.99
- Attendance to marks → 0.98
- Age to marks → 0.1

Output of df.corr()

Pair	Correlation
Study hrs to Study hrs	1
Study hrs to Attendance	0.98
Study hrs to Age	0.5
Study hrs to Marks	0.99
Attendance to Study hrs	0.98
Attendance to Attendance	1
Attendance to Age	0.6
Attendance to Marks	0.96

Pair	Correlation
Age to Study hrs	0.5
Age to Attendance	0.6
Age to Age	1
Age to Marks	0.3
Marks to Study hrs	0.99
Marks to Attendance	0.96
Marks to Age	0.3
Marks to Marks	1

Correlation Coefficient is applicable on **numerical inputs and outputs**.

CH- SQUARE METHOD:

• χ^2 :- It's applicable on Categorical inputs & outputs.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O :- observed points / original points
E :- Expected frequency
 $E = \frac{\text{Rows Total} \times \text{Columns Total}}{\text{Grand Total}}$

1) Consider following dataset

	Exercise diabetes		
	Yes	No	
low	30	10	→ 40
medium	10	20	→ 30
high	5	25	→ 30
	<u>45</u>	<u>55</u>	<u>100</u>

Expected frequency:

$$\text{Low (Yes)} = \frac{40 \times 45}{100} = 18$$

$$\text{Low (No)} = \frac{40 \times 55}{100} = 22$$

$$\text{Medium (Yes)} = \frac{30 \times 45}{100} = 13.5$$

$$\text{Medium (No)} = \frac{30 \times 55}{100} = 16.5$$

$$\text{High (Yes)} = \frac{30 \times 45}{100} = 13.5$$

$$\text{High (No)} = \frac{30 \times 55}{100} = 16.5$$

$$\begin{aligned} \frac{\sum (O - E)^2}{E} &= \frac{(30 - 18)^2}{18} + \frac{(10 - 22)^2}{22} + \frac{(10 - 13.5)^2}{13.5} + \frac{(20 - 16.5)^2}{16.5} \\ &\quad + \frac{(5 - 13.5)^2}{13.5} + \frac{(25 - 16.5)^2}{16.5} \end{aligned}$$

$$= 8 + 6.55 + 0.907 + 0.742 + 5.351 + 4.378$$

$$\chi^2 = 25.928$$

Compare this value with 'P' value i.e Z-statistic of 0.05, degree of freedom, $Z_{0.05, df} \Rightarrow Z_{0.05, 2} = 5.9$

$$\begin{aligned} df &= (\text{No of rows} - 1) \times (\text{No of Column} - 1) \\ &= (3 - 1) \times (2 - 1) \\ &= 2 \times 1 = 2 \end{aligned}$$

- If χ^2 value is high than $Z_{0.05, df}$ value, the feature is important
- If χ^2 value is less, the feature isn't important, hence it can be removed.

Since 25.9 is greater than 5.9, exercise is an important feature. If χ^2 value of any other feature is less than 5.9, it can be dropped.

→ Information gain, Fisher score are other methods which are used in Filter method for feature selection.

Whopper Method

1) Forward Selection:-

- Start with the Null Set & add features one by one.

Consider following dataset:-

→ St hours, Attendance, Age, Hobbies are independent variables & marks is dependent variable.

Calculate Statistic value (Accuracy, P-value & so on) for each & every feature.

Feature	Accuracy
St hours	80%
Attendance	70%
Age	50%
Hobbies	60%

From the above select the best feature i.e Study hours. included in the feature.

Repeat the above process till the Number of features are reached (2 best / 3 best features ...)

{ Study hours } → 80% → accuracy

{ Study hours, Attendance } → 90%

{ Study hours, Age } → 50%

{ Study hours, hobbies } → 60%

{ Study hours, Attendance, Age } →

{ St hours, Attendance, Hobbies } →

Features ↑
(1, 2, 3 combinations)

The two best features are St hours & Attendance.

2) Backward Elimination :- (Performance of the model)

- Select all the features & find the statistic value (Accuracy).
- Remove the features one by one & check the accuracy. Repeat the ab drop the one which doesn't effect the accuracy.
- Repeat the above process till maximum number of features is achieved (2/3 best features).

{ St hours, Attendance, Age, Hobbies } $\rightarrow$ 90%

{ St hours, Attendance, Hobbies } $\rightarrow$ 95%

{ St hours, Attendance } $\rightarrow$ 85%

3) Recursive Feature Elimination :-

- Train the model, Calculate feature Importance Score ^{generated by model}.
- Remove the least scored feature.
- Retrain the model with remaining features & Calculate the feature importance score.
- Repeat the above steps.

Feature	Score
① Study hours	0.65
Attendance	0.55
Age	0.35
Hobbies	0.32

$\nearrow$
drop hobbies

Feature	Score
② St hours	0.652
Attendance	0.5549
Age	0.3529

$\nearrow$ drop Age

Feature	Score
③ St hours	0.60
Attendance	0.50

Embedded Method

1) Lasso Regularisation:- This method adds a penalty which will zero minimum coefficients of the regression equation.

The usual linear regression considers all the coefficients whereas Lasso regularisation / regression doesn't consider the minimum coefficients.

$$\text{marks} = 0.9 \text{ Study hours} + 0.7 \text{ Attendance} + 0.2 \text{ Hobbies} + 0.1 \text{ Age} + c$$

In above eqn, in case of multiple linear regression all the features are considered though hobbies & Age are not important.

Lasso Regularisation will impose a penalty where least coefficients are zeroed.

$$\text{Marks} = 0.9 \text{ Study hours} + 0.7 \text{ Attendance} + c.$$

2.6 Dimensionality Reduction Techniques:

Dimensionality Reduction is reducing the number of features (variables) in a dataset while retaining **maximum useful information**.

Dimensionality reduction techniques: Dimensionality reduction technique can be defined as, "It is a way of converting the higher dimensions dataset into lesser dimensions dataset ensuring that it provides similar information." These techniques are widely used in machine learning for obtaining a better fit predictive model while solving the classification and regression problems.

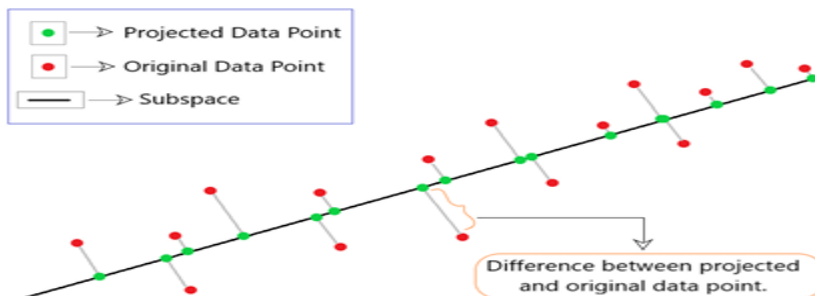
Benefits of applying Dimensionality Reduction:

- By reducing the dimensions of the features, the space required to store the dataset also gets reduced.
- Less Computation training time is required for reduced dimensions of features.
- Reduced dimensions of features of the dataset help in visualizing the data quickly.
- It removes the redundant features (if present) by taking care of multicollinearity.
- Avoids the **curse of dimensionality**.

Two classical approaches are **Projections** and **Eigenvalue Decomposition (EVD)**.

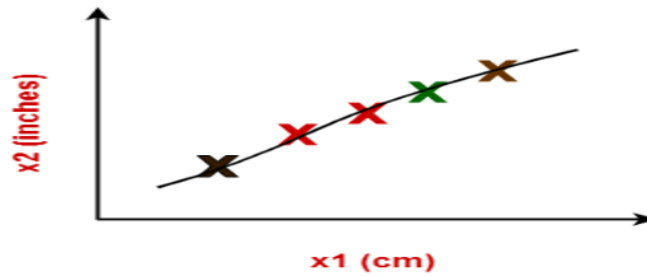
Types of Projection in Dimensionality Reduction

Projection is the process of mapping high-dimensional data onto a lower-dimensional subspace.



The following graph shows two dimensions x_1 and x_2 .

- x_1 represents the measurement of several objects in cm.
- x_2 represents the measurement of several objects in inches.



In machine learning,

- Using both these dimensions convey similar information.
- Also, they introduce a lot of noise in the system.
- So, it is better to use just one dimension.
-

Using dimension reduction techniques-

- We convert the dimensions of data from 2 dimensions (x_1 and x_2) to 1 dimension (z_1).
- It makes the data relatively easier to explain.



We mainly discuss 3 types of projection in the context of dimensionality reduction:

- ✧ Simple (or Feature Selection) Projection
- ✧ Linear Projection
- ✧ Optimal Projection (Data-driven, e.g., PCA)

2.6.1.1 Simple Projection (Feature Selection)

The simplest form of projection: just ignore some features and keep others. It

assumes that not all dimensions are equally important.

Example: If you have height, weight, and shoe size → you may just use height & weight and drop shoe size.

Mathematically,

If data point $x = (x_1, x_2, x_3)$,
a projection to 2D can be:

$$y = (x_1, x_2)$$

Just drop the 3rd coordinate.

Dataset in 3D:

(2,4,5),(3,6,8),(4,5,9)

Projecting onto (x, y):

(2,4), (3,6),(4,5)

Linear Projection

Instead of simply dropping dimensions, we create new features as linear combinations of original features.

Allows mixing information from multiple dimensions.

If data point $x = (x_1, x_2, x_3)$,
a linear projection to 2D can be:

$$y_1 = a_1x_1 + a_2x_2 + a_3x_3$$

$$y_2 = b_1x_1 + b_2x_2 + b_3x_3$$

Here a_i, b_i are weights defining the new axes.

$$y_1 = 0.8x_1 + 0.2x_2 + 0.0x_3, \quad y_2 = 0.5x_2 + 0.5x_3$$

Data point (2, 4, 6):

$$y_1 = 0.8(2) + 0.2(4) = 1.6 + 0.8 = 2.4$$

$$y_2 = 0.5(4) + 0.5(6) = 2 + 3 = 5$$

So, $(2, 4, 6) \rightarrow (2.4, 5)$.

This method preserves more info than just dropping coordinates.

Optimal Projection (Data-driven, e.g., PCA)

Principal Component Analysis (PCA) & Eigen value decomposition:

Principal Component Analysis (PCA) is a powerful technique used in data analysis, particularly for reducing the dimensionality of datasets while preserving crucial information. It does this by transforming the original variables into a set of new, uncorrelated variables called principal components.

- Principal Component Analysis is a well-known dimension reduction technique.
- It transforms the variables into a new set of variables called as principal components.
- These principal components are linear combination of original variables and are orthogonal.
- The first principal component accounts for most of the possible variation of original data.
- The second principal component does its best to capture the variance in the data.
- There can be only two principal components for a two-dimensional data set.

The steps involved in PCA Algorithm are as follows –

Step-01: Get data.

Step-02: Compute the mean vector (μ).

Step-03: Subtract mean from the given data.

Step-04: Calculate the covariance matrix.

Step-05: Calculate the eigen vectors and eigen values of the covariance matrix.

Step-06: Choosing components and forming a feature vector.

Step-07: Deriving the new data set.

PROBLEM: Given data = { 2, 3, 4, 5, 6, 7 ; 1, 5, 3, 6, 7, 8 }. Compute the principal component using PCA Algorithm.

OR

Consider the two dimensional patterns (2, 1), (3, 5), (4, 3), (5, 6), (6, 7), (7, 8). Compute the principal component using PCA Algorithm.

OR

Compute the principal component of following data-

CLASS 1 = X = 2, 3, 4 ; Y = 1, 5, 3

CLASS 2 = X = 5, 6, 7 ; Y = 6, 7, 8

Step-01: Get data. The given feature vectors are-

x1 = (2, 1)

x2 = (3, 5)

x3 = (4, 3)

x4 = (5, 6)

x5 = (6, 7)

x6 = (7, 8)

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Step-02: Calculate the mean vector (μ).

Mean vector (μ) = $((2 + 3 + 4 + 5 + 6 + 7) / 6, (1 + 5 + 3 + 6 + 7 + 8) / 6) = (4.5, 5)$

$$\text{Mean vector } (\mu) = \begin{bmatrix} 4.5 \\ 5 \end{bmatrix}$$

Step-03: Subtract mean vector (μ) from the given feature vectors.

- $x1 - \mu = (2 - 4.5, 1 - 5) = (-2.5, -4)$

- $x_2 - \mu = (3 - 4.5, 5 - 5) = (-1.5, 0)$
- $x_3 - \mu = (4 - 4.5, 3 - 5) = (-0.5, -2)$
- $x_4 - \mu = (5 - 4.5, 6 - 5) = (0.5, 1)$
- $x_5 - \mu = (6 - 4.5, 7 - 5) = (1.5, 2)$
- $x_6 - \mu = (7 - 4.5, 8 - 5) = (2.5, 3)$

Feature vectors (x_i) after subtracting mean vector (μ) are-

$$\begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 2.5 \\ 3 \end{bmatrix}$$

Step-04: Calculate the covariance matrix. Covariance matrix is given by-

$$\text{Covariance Matrix} = \frac{\sum (x_i - \mu)(x_i - \mu)^t}{n}$$

$$m_1 = (x_1 - \mu)(x_1 - \mu)^t = \begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -2.5 & -4 \end{bmatrix} = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix}$$

$$m_2 = (x_2 - \mu)(x_2 - \mu)^t = \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -1.5 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix}$$

$$m_3 = (x_3 - \mu)(x_3 - \mu)^t = \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} -0.5 & -2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 \\ 1 & 4 \end{bmatrix}$$

$$m_4 = (x_4 - \mu)(x_4 - \mu)^t = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$m_5 = (x_5 - \mu)(x_5 - \mu)^t = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 1.5 & 2 \end{bmatrix} = \begin{bmatrix} 2.25 & 3 \\ 3 & 4 \end{bmatrix}$$

$$m_6 = (x_6 - \mu)(x_6 - \mu)^t = \begin{bmatrix} 2.5 \\ 3 \end{bmatrix} \begin{bmatrix} 2.5 & 3 \end{bmatrix} = \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

Covariance matrix = $(m_1 + m_2 + m_3 + m_4 + m_5 + m_6) / 6$
On adding the above matrices and dividing by 6, we get-

$$\text{Covariance Matrix} = \frac{1}{6} \begin{bmatrix} 17.5 & 22 \\ 22 & 34 \end{bmatrix}$$

$$\text{Covariance Matrix} = \begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix}$$

Step-05: Calculate the eigen values and eigen vectors of the covariance matrix.

λ is an eigen value for a matrix M if it is a solution of the characteristic equation $|M - \lambda I| = 0$.

So, we have-

$$\begin{vmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2.92 - \lambda & 3.67 \\ 3.67 & 5.67 - \lambda \end{vmatrix} = 0$$

From here, $(2.92 - \lambda)(5.67 - \lambda) - (3.67 \times 3.67) = 0$

$$16.56 - 2.92\lambda - 5.67\lambda + \lambda^2 - 13.47 = 0$$

$$\lambda^2 - 8.59\lambda + 3.09 = 0$$

Solving this quadratic equation,

we get $\lambda = 8.22, 0.38$

Thus, two eigen values are $\lambda_1 = 8.22$ and $\lambda_2 = 0.38$.

the second eigen value is very small compared to the first eigen value.

So, the second eigen vector can be left out.

Eigen vector corresponding to the greatest eigen value is the principal component for the given data set.

So we find the eigen vector corresponding to eigen value λ_1 .

We use the following equation to find the eigen vector -

$$MX = \lambda X \text{ where -}$$

- M = Covariance Matrix
- X = Eigen vector
- λ = Eigen value

Substituting the values in the above equation, we get-

$$\begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = 8.22 \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

Solving these,
we get-

$$2.92X_1 + 3.67X_2 = 8.22X_1$$

$$3.67X_1 + 5.67X_2 = 8.22X_2$$

On simplification,

we get -

$$5.3X_1 = 3.67X_2 \dots\dots\dots(1)$$

$$3.67X_1 = 2.55X_2 \dots\dots\dots(2)$$

From (1) and (2), $X_1 = 0.69X_2$ From (2), the eigen vector is -

$$\text{Eigen Vector : } \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 2.55 \\ 3.67 \end{bmatrix}$$

Thus, principal component for the given data set is-

$$\text{Principal Component : } \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 2.55 \\ 3.67 \end{bmatrix}$$

The above can be further normalized:

$$\frac{X_1}{\sqrt{X_1^2 + X_2^2}}, \frac{X_2}{\sqrt{X_1^2 + X_2^2}}$$

$$X_1 = 2.55, X_2 = 3.67$$

$$\frac{2.55}{\sqrt{2.55^2 + 3.67^2}} = 0.570, \frac{3.67}{\sqrt{2.55^2 + 3.67^2}} = 0.821$$

The normalized eigen vectors are 0.570 & 0.821
 The above are the principal Components.

⑦ $Y = 0.57X_1 + 0.82X_2$

∴ Y is the new feature

X_1	X_2	Y
2	1	1.96
3	5	5.81
4	3	4.74
5	6	7.77
6	7	9.16
7	8	10.55

Consider the following data points

(1, 2)	1	2
(2, 3)	2	3
(3, 5)	3	5
(4, 6)	4	6
(5, 8)	5	8
	3	hr

(a). Find Covariance Matrix for the above.

$$\frac{\sum (X_i - \mu)(X_i - \mu)^T}{n}$$

$$\mu = \begin{bmatrix} 3 \\ 4.8 \end{bmatrix}$$

$$\text{Step 1: } x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1-3 \\ 2-4.8 \end{bmatrix} = \begin{bmatrix} -2 \\ -2.8 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1.8 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.2 \end{bmatrix}$$

$$x_4 = \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.2 \end{bmatrix}$$

$$x_5 = \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ 3.2 \end{bmatrix}$$

$$\text{Step 2: } x_1 = \begin{bmatrix} -2 \\ -2.8 \end{bmatrix} \begin{bmatrix} -2 & -2.8 \end{bmatrix} = \begin{bmatrix} 4 & 5.6 \\ 5.6 & 7.84 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} -1 \\ -1.8 \end{bmatrix} \begin{bmatrix} -1 & -1.8 \end{bmatrix} = \begin{bmatrix} 1 & 1.8 \\ 1.8 & 3.24 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} \begin{bmatrix} 0 & 0.2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0.04 \end{bmatrix}$$

$$x_4 = \begin{bmatrix} -4 \\ -6 \end{bmatrix} \begin{bmatrix} -4 & -6 \end{bmatrix} = \begin{bmatrix} 16 & 24 \\ 24 & 36 \end{bmatrix}$$

$$x_5 = \begin{bmatrix} 2 \\ 3.2 \end{bmatrix} \begin{bmatrix} 2 & 3.2 \end{bmatrix} = \begin{bmatrix} 4 & 6.4 \\ 6.4 & 10.24 \end{bmatrix}$$

$$\frac{\sum (x_i - \mu) (x_i - \mu)^T}{n} = \frac{1}{5} \begin{bmatrix} 10 & 15 \\ 15 & 22.8 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 \\ 3 & 4.56 \end{bmatrix}$$

$$\text{Covariance Matrix} = \begin{bmatrix} \text{Variance}(x_1) & \text{Covariance}(x_1, x_2) \\ \text{Covariance}(x_1, x_2) & \text{Variance}(x_2) \end{bmatrix}$$

The subtraction of mean value from the original data points will make the data centered.

Step 3:- $|M - \lambda I| = 0$

$$= \begin{vmatrix} 2 & 3 \\ 3 & 4.56 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 - \lambda & 3 \\ 3 & 4.56 - \lambda \end{vmatrix} = 0$$

$$2 - \lambda (4.56 - \lambda) = 0 \quad - 3(3) = 0$$

$$9.12 - 2\lambda - 4.56\lambda - \lambda^2 - 9 = 0$$

$$\lambda^2 - 6.56\lambda + 0.12 = 0$$

$$\lambda_1 = 0.12$$

$$\lambda_2 = 6.54$$

$$\lambda_2 = 0.018$$

Ignore the least value.

$$\therefore \lambda = 6.54$$

Eigen Vector :- $MX = \lambda X$

$$\begin{bmatrix} 2 & 3 \\ 3 & 4.56 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 6.54 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= 2x_1 + 3x_2 = 6.54x_1 \quad \Rightarrow 3x_2 = 6.54x_1 - 2x_1$$

$$= 3x_1 + 4.56x_2 = 6.54x_2$$

$$3x_2 = 4.54x_1$$

$$3x_1 = 6.54x_2 - 4.56x_2$$

$$3x_1 = 1.98x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1.98 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4.54 \end{bmatrix}$$

Normalized:- $X_1 = 1.98$ $X_2 = 3$

$$\frac{X_1}{\sqrt{X_1^2 + X_2^2}}, \frac{X_2}{\sqrt{X_1^2 + X_2^2}}$$

$$= \frac{1.98}{\sqrt{1.98^2 + 3^2}}, \frac{3}{\sqrt{1.98^2 + 3^2}}$$

$$= 0.55 \quad = 0.83$$